

Equipment Manual

IIT X-Ray Diffractometer

Equipment ID: 31

DEMO / TEST DOCUMENT

Equipment	IIT X-Ray Diffractometer
Equipment ID	31
Document type	Equipment operating manual
Intended use	Laboratory resource management and demonstration

1. Purpose and Overview

This document provides general operating and handling guidance for IIT X-Ray Diffractometer. It is prepared as a demonstration document for the Lab Resource Utilization Platform. Users should follow the laboratory's approved procedures and the manufacturer's official instructions for actual operation.

2. Before Use

Verify that the equipment is assigned to the correct laboratory, is visibly clean and intact, and is marked as available for use. Check required accessories, power connections, environmental conditions, and any displayed warnings before starting.

3. Basic Operating Procedure

1) Confirm the equipment identity and booking. 2) Inspect the equipment and work area. 3) Power on using the approved startup procedure. 4) Configure the required operating parameters according to the laboratory procedure. 5) Perform the intended measurement, preparation, processing, or storage task. 6) Record relevant observations and usage details in the laboratory record.

4. Safety Precautions

Only trained and authorized personnel should operate laboratory equipment. Use required personal protective equipment. Keep liquids, samples, cables, and other materials away from unsafe areas. Never bypass safety interlocks or alarms. Stop operation and report abnormal noise, smell, temperature, leakage, electrical behavior, or error messages.

5. Cleaning and Routine Care

After use, return controls to the approved safe state, remove samples and consumables, clean accessible surfaces using the laboratory-approved method, and leave the workspace ready for the next user. Do not open covers or perform internal maintenance unless authorized.

6. Troubleshooting

If the equipment does not start, check the approved power and connection points. If an error appears, record the exact message and follow the laboratory troubleshooting procedure. Do not repeatedly restart equipment showing a persistent fault. Contact the responsible lab manager or maintenance team when required.

7. Shutdown Procedure

Complete or safely terminate the current operation, save required results, remove materials, place the equipment in its approved standby or shutdown state, and clean the work area. Record any fault, unusual event, or maintenance requirement before leaving.

8. Maintenance and Calibration

Maintenance and calibration must follow the laboratory schedule and approved procedures. Calibration dates shown in the resource platform are administrative records and should be verified against the laboratory's official calibration program.

9. Record Keeping

Record user, date, booking reference where applicable, relevant operating conditions, abnormal observations, cleaning status, and maintenance/calibration observations according to laboratory policy.

Important: This is a generated demonstration manual. It is not a manufacturer-issued manual and must not replace approved laboratory procedures or manufacturer documentation.